AIM :

To perform the operation of filing, marking and punching

Tools required:

1. Bench vice
2. Try square
3. Hacksaw frame
4. Scriber
5. Steel rule
6. Punches
7. Flat file-rough
8. Flat file-medium
9. Flat file-smooth
10. Triangular file
11. Hammer
12. Divider
13. Chalk paste
14. Vernier caliper
15. Surface plate

PROCEDURE

1. The design of the given model is studied completely to understand the features like its size and shape.
2. The given piece of metal is checked for size whether it is sufficient for the design.
3. Hand tools and measuring instruments are selected suitably to perform the required operations.
4. The piece of metal (Mild Steel) is held in the bench vice and two adjacent sides are filed for squareness using a flat file. These two sides are considered prime sides.
5. The remaining two sides are also filed for squareness (The angle between adjacent sides=90 degrees). The same is checked with help of a try- square .
6. The piece of work is checked for 90° on all four sides.
7. A thin layer of chalk paste is applied on the flat surface of the metal piece .
8. Chalk is allowed to dry.
9. The given design is scribed on the chalked surface using steel rule, divider and scriber.
10. Punch marks are made at required points on the surface using dot punch and a hammer.
11. Punch marks are made by keeping the punch inclined 60° so that they are filed off later.